

Exploratory Analysis of Real-World Radiosonde Profiles

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1 Introduction

This project uses real-world radiosonde observations from the **University of Wyoming Atmospheric Science Radiosonde Archive** to explore how atmospheric pressure, temperature, and wind vary with height. Soundings from Singapore are used as a tropical reference and compared with observations from Ulaanbaatar, Mongolia, which provides a contrasting colder and drier continental environment.

The analysis focuses on vertical pressure structure, seasonal pressure–temperature profiles, layer-average lapse rates, and wind-speed profiles. Observations from January and July 2021 are used to examine how the two atmospheric columns differ under contrasting seasonal conditions. The results describe the selected soundings rather than long-term climatological averages.

2 Data and Workflow

The radiosonde text files were loaded into Python as structured numerical tables containing pressure, geopotential height, temperature, moisture, and wind variables. The workflow used for the results reported here was:

1. load and validate the radiosonde observations for Singapore and Ulaanbaatar at 00Z and 12Z;
2. extract pressure–height profiles for direct comparison between stations and launch times;
3. compare pressure–temperature profiles for the January and July 12Z soundings;
4. interpolate temperature with respect to recorded height at fixed altitude boundaries and calculate layer-average lapse rates; and
5. compare wind-speed profiles over the observed pressure levels.

The accompanying notebook contains the data-loading and plotting workflow used to produce the reported tables and figures.

3 Pressure–Height Structure

3.1 Observed pressure levels

The pressure–height observations used in the first comparison are summarized in **Table 1** and **Table 2**. Ulaanbaatar is located at a substantially higher station elevation, approximately 1306 m.

Table 1: Singapore Station: 7th July 2021

Pressure (mb)	Height (m) 0Z	Height (m) 12Z
1000	96	88
925	779	780
850	1511	1510
700	3150	3152
500	5860	5870
400	7590	7600
300	9700	9710
200	12450	12460
100	16580	16590

Table 2: Ulaanbaatar Station: 7th July 2021

Pressure (mb)	Height (m) 00Z	Height (m) 12Z
850	1517	1483
700	3149	3148
500	5820	5840
400	7480	7510
300	9520	9550
200	12160	12200
100	16590	16610

3.2 Pressure-profile visualization

The measured pressure profiles are shown in **Figure 1**. Atmospheric pressure decreases non-linearly with increasing height. This reflects the reduction in the mass of air above a given level as altitude increases, producing progressively lower pressure higher in the atmosphere.

Small differences are present between the two launch times, but they are difficult to distinguish at the scale of the July plot. The Singapore and Ulaanbaatar profiles are also relatively close at the selected pressure levels in July. To examine whether the station contrast changes under different seasonal conditions, the same comparison is repeated using January observations.

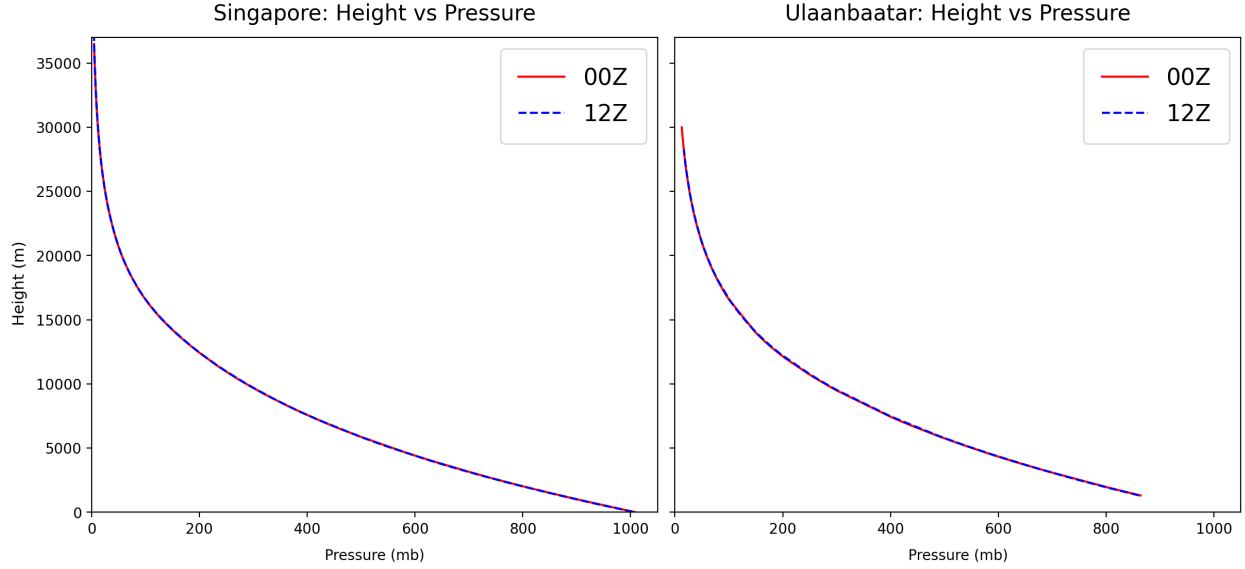


Figure 1: Singapore & Ulaanbaatar Pressure Profile

3.3 January pressure-level comparison

As shown in **Table 3** and **Table 4**, the January soundings show a noticeably larger geopotential-height difference between Singapore and Ulaanbaatar, especially at upper pressure levels. For example, the 500 mb level is located at approximately 5850 m over Singapore, compared with about 5270–5330 m over Ulaanbaatar.

Table 3: Singapore Station: 7th January 2021

Pressure (mb)	Height (m) 00Z	Height (m) 12Z
1000	83	68
925	764	750
850	1494	1482
700	3133	3127
500	5850	5850
400	7570	7570
300	9700	9690
200	12460	12440
100	16560	16540

Table 4: Ulaanbaatar Station: 7th January 2021

Pressure (mb)	Height (m) 00Z	Height (m) 12Z
850	1525	1535
700	2918	2940
500	5270	5330
400	6760	6850
300	8580	8700
200	11100	11210
100	15421	15460

This difference can be explained by the hypsometric equation:

$$z_2 - z_1 = \frac{R_d \overline{T}_v}{g} \ln \left(\frac{p_1}{p_2} \right) \quad (1)$$

where $z_2 - z_1$ is the thickness of the atmospheric layer, R_d is the specific gas constant for dry air, $\overline{T}_v$ is the mean virtual temperature of the layer, and p_1 and p_2 are the lower and upper pressure levels.

The recorded mixing ratios indicate that the Singapore sounding was moister than the Ulaanbaatar sounding. Using the approximation $T_v \approx T(1 + 0.61r_v)$, where r_v is the mixing ratio, greater moisture increases virtual temperature for a given actual temperature. Singapore’s warmer and moister air therefore corresponds to a higher mean virtual temperature, while Ulaanbaatar’s colder and drier air corresponds to a lower value.

According to the hypsometric equation, a higher mean virtual temperature corresponds to a greater thickness between two fixed pressure levels. The Singapore atmospheric column is therefore thicker between comparable pressure surfaces than the colder Ulaanbaatar column in the January soundings. This difference is shown more clearly in **Figure 2**.

4 Singapore Pressure–Temperature Profiles

The Singapore 12Z soundings from 7 January and 7 July 2021 are compared to examine how the vertical temperature structure differs between the two selected dates. The pressure–temperature values used for this comparison are summarized in **Table 5**.

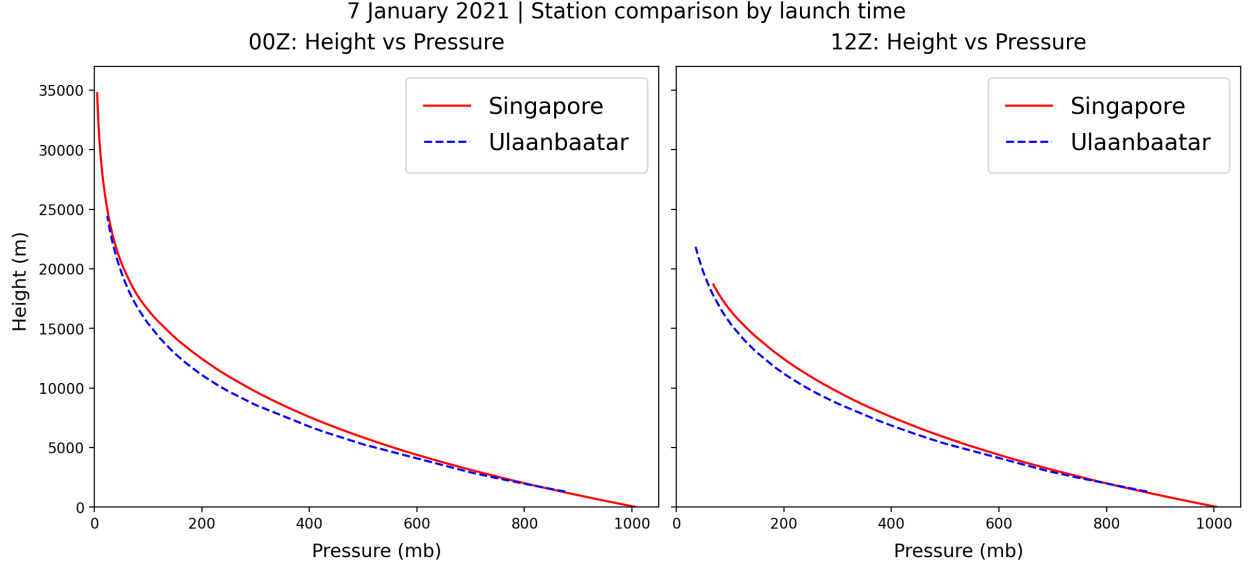


Figure 2: Singapore & Ulaanbaatar Pressure Profile Comparison in 7th January 2021

Pressure (mb)	January ($^{\circ}\text{C}$)	July ($^{\circ}\text{C}$)
1000	24.4	28.6
925	21.6	22.8
850	18.4	18.2
700	11.2	9.2
500	-5.1	-4.3
400	-15.1	-16.1
300	-29.5	-30.1
200	-52.9	-53.3
100	-78.9	-69.9

Table 5: Singapore Temperature and Pressure at 12Z on 7 January and 7 July 2021.

Pressure and temperature both generally decrease with increasing altitude through much of the observed troposphere. Near the surface, the January and July soundings differ by approximately 4.2°C , while the temperature difference becomes much smaller by the 850 mb level.

The cooler January near-surface conditions are consistent with Northeast Monsoon influence, during which cold surges associated with the Siberian High can reach the near-equatorial region and contribute to cooler, cloudier, and wetter conditions. The full Singapore pressure–temperature profiles are shown in **Figure 3**.

Another feature visible in **Figure 3** is a temperature inversion near the 100 mb level, corresponding to the transition from the tropopause region into the lower stratosphere.

To quantify the vertical temperature gradient, layer-average lapse rates were calculated using temperature interpolated at fixed altitude boundaries. The January and July 12Z results for Singapore are shown in **Table 6** and **Table 7**.

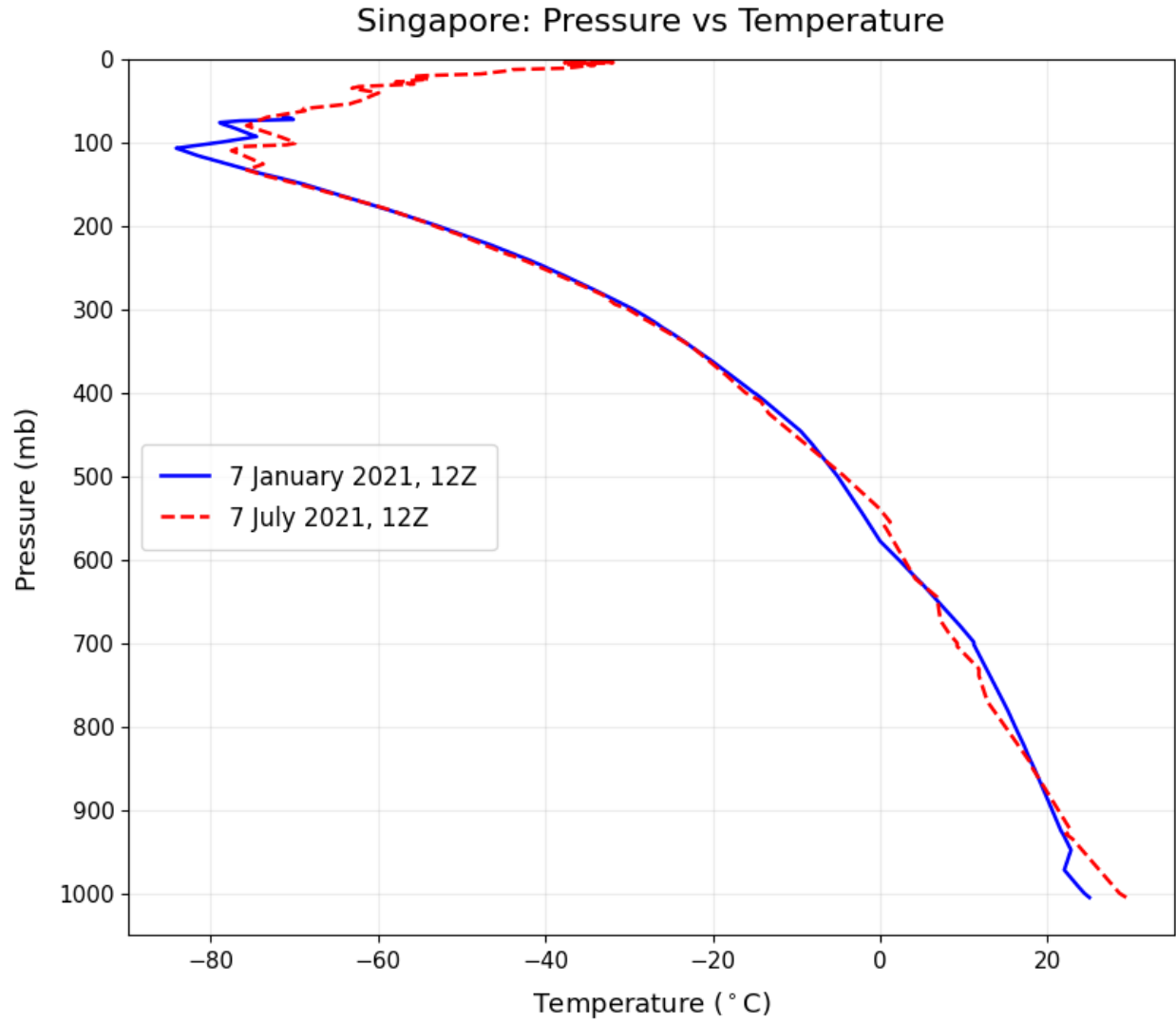


Figure 3: Singapore Pressure-Temperature Profile

Altitude (m)	Temperature (°C)	Lapse Rate (°C / km)
16	25.00	—
1000	20.49	4.51
2000	16.10	4.39
3000	11.66	4.44
4000	5.03	6.63
5000	-1.36	6.39
6000	-5.84	4.48
7000	-11.24	5.40
8000	-18.03	6.79
9000	-24.79	6.76
10000	-32.07	7.28

Table 6: Singapore temperatures lapse rate at 12Z on 7 January 2021.

Altitude (m)	Temperature ($^{\circ}\text{C}$)	Lapse Rate ($^{\circ}\text{C} / \text{km}$)
16	29.40	–
1000	21.30	8.10
2000	15.02	6.28
3000	9.92	5.10
4000	5.06	4.86
5000	0.66	4.40
6000	-5.30	5.96
7000	-12.44	7.14
8000	-18.50	6.06
9000	-24.87	6.38
10000	-32.52	7.64

Table 7: Singapore temperatures lapse rate at 12Z on 7 July 2021.

The calculated lapse rate is not uniform with altitude and differs between the two selected soundings. In the lower atmosphere, January shows lapse rates of about $4.5^{\circ}\text{C}/\text{km}$, closer to typical moist-adiabatic values, while July shows a much larger near-surface lapse rate of about $8.1^{\circ}\text{C}/\text{km}$. At higher altitudes, both profiles continue to vary from layer to layer. The smaller lower-atmospheric lapse rates in the January sounding are consistent with stronger moist-process influence and latent-heat release under humid and cloudy Northeast Monsoon conditions.

5 Ulaanbaatar Pressure–Temperature Profiles

The same approach is applied to the Ulaanbaatar soundings to examine the vertical temperature structure in a contrasting continental climate. Layer-average lapse rates for the January and July 12Z profiles are shown in **Table 8** and **Table 9**.

Altitude (m)	Temperature ($^{\circ}\text{C}$)	Lapse Rate ($^{\circ}\text{C}/\text{km}$)
1306	-22.90	–
2000	-25.23	3.36
3000	-25.86	0.62
4000	-29.81	3.95
5000	-34.69	4.88
6000	-39.87	5.18
7000	-47.72	7.85
8000	-55.12	7.40
9000	-61.18	6.06
10000	-60.83	-0.36

Table 8: Ulaanbaatar temperatures and layer-average lapse rates at 12Z on 7 January 2021.

Altitude (m)	Temperature ($^{\circ}\text{C}$)	Lapse Rate ($^{\circ}\text{C}/\text{km}$)
1306	27.60	—
2000	19.99	10.97
3000	10.61	9.38
4000	3.56	7.05
5000	-3.63	7.19
6000	-9.15	5.52
7000	-17.01	7.86
8000	-24.97	7.96
9000	-33.03	8.06
10000	-41.08	8.05

Table 9: Ulaanbaatar temperatures and layer-average lapse rates at 12Z on 7 July 2021.

A clear feature of the Ulaanbaatar profiles is the stronger contrast between the two selected seasons in the lower atmosphere. In January, the 2000–3000 m layer has a lapse rate of only $0.62^{\circ}\text{C}/\text{km}$, indicating a relatively stable layer. A slight negative lapse rate of $-0.36^{\circ}\text{C}/\text{km}$ appears near 10,000 m, indicating layer-average warming with height. In July, the near-surface lapse rate reaches $10.97^{\circ}\text{C}/\text{km}$, close to the theoretical dry-adiabatic value and consistent with the station’s comparatively dry continental conditions.

Both stations show non-uniform lapse rates, but the contrast between the January and July profiles is much stronger for Ulaanbaatar. The January sounding reflects a much colder and more stable lower atmospheric column, consistent with strong wintertime cooling and anticyclonic conditions that suppress vertical mixing.

Figure 4 further illustrates the stronger January–July contrast in Ulaanbaatar. The January profile is substantially colder than the Singapore profile through much of the troposphere, while the two stations are closer in July. This supports the lapse-rate comparison: the selected Ulaanbaatar soundings range from a very stable winter lower atmosphere to a much steeper, near dry-adiabatic lower-level temperature gradient in summer. The colder January column also connects with the pressure–height analysis, because lower mean temperature produces a smaller vertical thickness between fixed pressure levels.

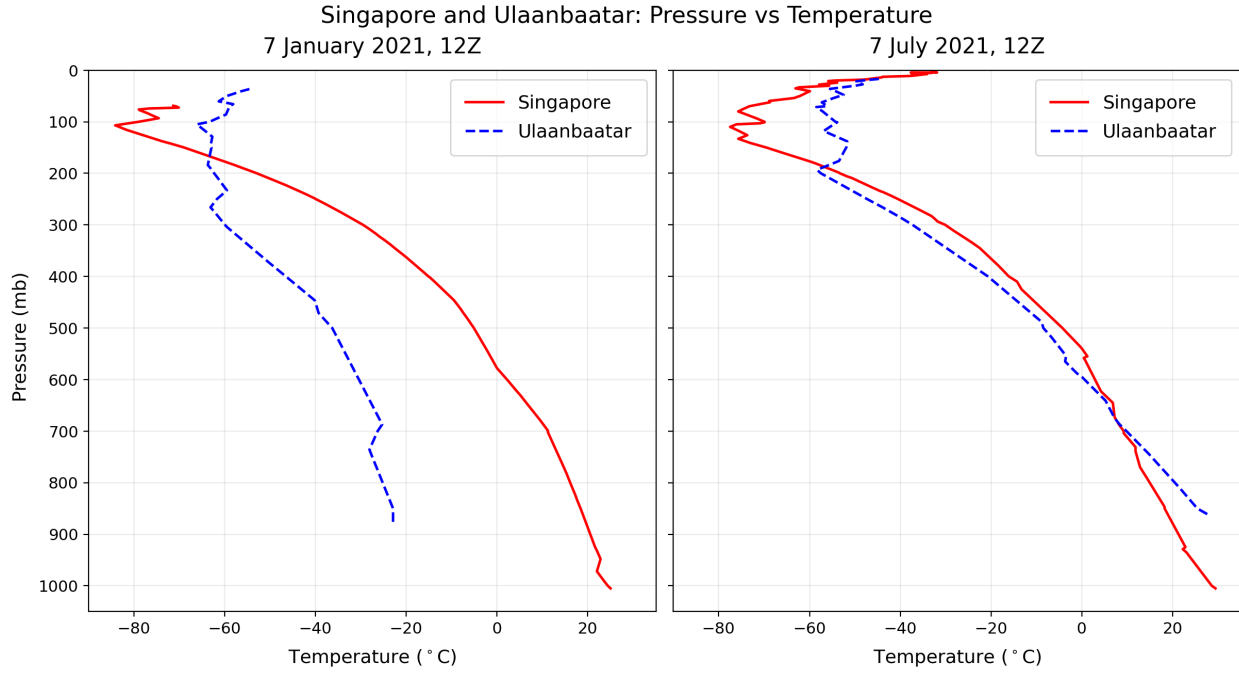


Figure 4: Ulaanbaatar Pressure-Temperature Profile

6 Wind-Speed Profiles

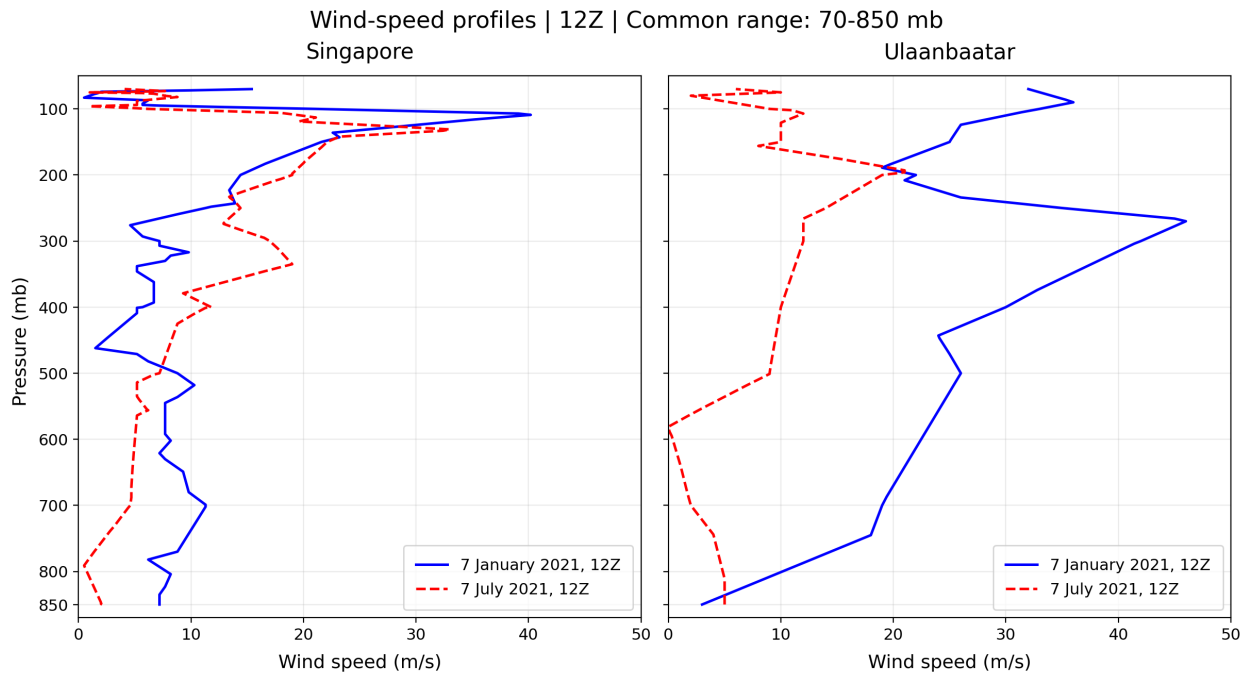


Figure 5: Wind-speed profiles for Singapore and Ulaanbaatar

6.1 Singapore

Figure 5 shows that wind speed varies considerably with pressure level rather than changing uniformly with altitude. January winds are stronger at the lower reference levels, whereas July winds are stronger at several levels between 400 and 200 mb. The profiles therefore highlight substantial vertical variability in wind speed within both soundings.

6.2 Ulaanbaatar

Ulaanbaatar shows a much stronger January–July contrast than Singapore. Across much of the observed profile, January wind speeds are higher than those in July. This pattern is consistent with stronger wintertime horizontal temperature gradients over continental Asia and the associated vertical wind-shear relationship described by thermal-wind balance.

7 Summary and Conclusion

This exploratory analysis demonstrates how real radiosonde observations can be used to examine differences in atmospheric structure between a tropical station and a colder continental station. Across the selected Singapore and Ulaanbaatar soundings, clear contrasts appear in pressure-level height, temperature structure, lapse rate, and wind speed. The observed features are consistent with physical concepts including the hypsometric relationship, anticyclonic winter conditions associated with the Siberian High, temperature inversions, tropopause behaviour, and dry- versus moist-adiabatic lapse rates.

The exercise also illustrates how pressure, temperature, moisture, stability, and wind are connected within a vertical atmospheric column. Working directly with observed sounding data provided practical experience in loading and validating environmental data, constructing vertical-profile visualizations, calculating layer diagnostics, and relating measured structure to atmospheric theory.